
Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

Foundation models for biology are increasingly probed with interpretability methods that identify correlational structure. Examples include an attention head aligned with a known biological signal or a scoring proxy correlated with real experimental labels. This structure is often treated as evidence that a targeted causal intervention, such as activation steering or attribution-guided editing, will work as intended. We test this assumption directly on a protein language model, ESM2-650M, using difference-of-means activation steering, a causal intervention already shown to work for one property (thermostability). Under a pre-registered six-criterion protocol testing five novel target properties, we find no positive relationship between proxy validity and causal steering effect: our best-validated proxy (binding affinity, $r = 0.80$ held-out, confirmed by a weight-shuffle null control) steers nothing, while a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated causal effect. An independent, component-level mechanistic-interpretability result shows the identical pattern one level down: attention heads ranked by correlation with real 3D protein contacts show near-zero relationship to their causal ablation effect, confirmed via a full 480-head sweep. We further show that a single evaluation seed can produce a misleading verdict on a specific robustness criterion even when the underlying causal effect is stable. We also introduce a diagnostic based on steering-vector cosine similarity across targets. It explains an otherwise unexplained result as a geometric artifact of a different, validated causal direction rather than independent evidence. These findings argue for validating explanation methods against downstream causal interventions, not treating correlational salience as self-certifying.

1 Introduction

A foundation model’s internal correlational structure includes which attention heads track a known signal and which features predict a real label. This structure is frequently used as the basis for a claimed causal intervention: ablate this head because it looks important, or steer along this direction because it correlates with the target property. Whether that correlational salience actually predicts the intervention’s causal effect is rarely tested directly. We test it on ESM2-650M, a protein language model, using difference-of-means activation steering [Huang et al., 2025], a method already validated to causally move thermostability. It adds a direction to the model during generation, built from the residual-stream activation difference between real high- and low-property sequence groups.

We ask: does a scoring proxy’s correlation with real experimental labels predict whether steering toward it causally works? Across five novel target properties under one pre-registered protocol, the answer is no, and not weakly so. The best-validated proxy in our study ($r = 0.80$ held-out, for binding

Table 1: Five novel target properties. Proxy r is Pearson correlation against real labels on a held-out split, computed before any steering (causal-intervention) run.

Target	Dataset	Proxy r	Steering result	Verdict
Binding affinity	ProteinGym DMS (KRAS/DARPin)	0.80	flat null, all α	KILL
Catalytic activity	DLKcat (cross-protein)	0.22	PASS, 3/3 seeds	PASS
Intrinsic disorder	DisProt	0.45	PASS, 3/3 direction; 2/3 robustness	PASS*
Immunogenicity	IEDB (4-tier)	≤ 0.10	not run	KILL (pre-run)
Expression yield	eSol	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1, 2, 4, 6, and 2 of 3 seeds on criterion 3 (§3.1).

affinity) produces a flat null causal effect at every tested intervention strength, while a substantially weaker proxy ($r = 0.22$, catalytic activity) produces a clean, monotonic effect that replicates across three independent random seeds. This mirrors an independent, purely mechanistic-interpretability result in the same model family: Vig et al. [2021] identified an attention head whose attention pattern correlates $12.9\times$ with real 3D residue contacts, explicitly noting this was never tested causally; we ablate it directly and find an effect statistically indistinguishable from a low-correlation control, confirmed by a full sweep of all 480 heads in the model. At two levels, individual components and entire target properties, we find one lesson: correlational salience, however striking, is evidence about a model’s learned representations, not a certificate that an intervention built on it will causally succeed.

2 Method: A Pre-Registered Protocol for Causal Validation

Before testing any target property, we lock a six-criterion decision rule that is applied uniformly and computed automatically. Criterion 1 requires the real steering direction to beat a matched-norm random control via paired bootstrap with 10,000 resamples. Criterion 2 requires a monotonic dose-response across at least three strengths within a safe operating range (§4). Criterion 3 requires the effect to survive a residue-exclusion check that rules out compositional collapse. Criterion 4 requires validation of the scoring proxy against real labels before steering. This is the causal-structure analogue of validating an explanation method against ground truth before trusting its downstream use. Criterion 5 requires the result to beat the best existing technique when one exists, and criterion 6 requires at least 150 held-out sequences. A target failing any criterion is a KILL; one passing 1–2 but failing 3 or 4 is AMBIGUOUS.

3 Results: Explanatory Validity Does Not Predict Causal Success

Binding affinity. Using a ProteinGym deep-mutational-scanning assay of KRAS binding to a DARPin binder, we build a proxy from per-position mutational-sensitivity weights learned on the training split. This is the most rigorously validated explanation in our study: $r = 0.80$ held-out. A weight-shuffle null of $r = -0.001$ rules out a generic rather than position-specific signal. The proxy also generalizes to held-out positions, with $r = 0.54$ on positions never seen during weight fitting, and from single to double mutants, with $r = 0.70$. The causal intervention built from this explanation is a flat, unambiguous null at every tested strength, with zero degenerate sequences. This rules out generation collapse as an alternative account. A plausible mechanistic reason (§4): this dataset’s vector-building groups are single-backbone point mutants, giving the difference-of-means construction very little raw compositional signal to act on, independent of whether the underlying explanation is valid.

Catalytic activity. A glycine-minus-arginine compositional proxy on DLKcat (cross-protein enzyme turnover data), the weakest explanation of the four runnable targets ($r = 0.22$), matching the published activity-stability tradeoff in enzymology. This produces a clean, monotonic causal effect (Figure 2, left), significant at every safe-range α , surviving residue-exclusion, and replicated

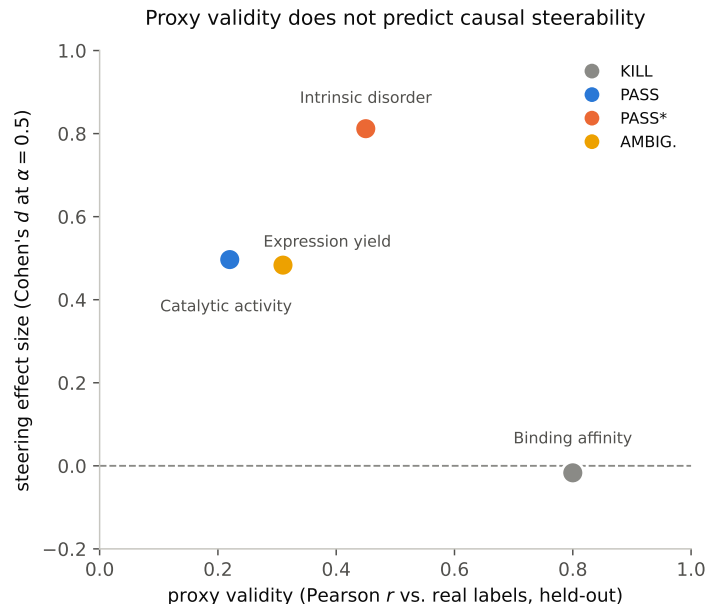


Figure 1: The four runnable targets, plotted as explanatory validity (proxy Pearson r against real labels) against causal steering effect size (Cohen’s d at $\alpha = 0.5$). The most trustworthy explanation (binding affinity, $r = 0.80$) yields the least causally useful intervention; a far weaker explanation ($r = 0.22$, catalytic activity) yields one of the two most causally effective interventions.

73 identically across three independent seeds. It is the one target in our sweep with fully unanimous
 74 seed robustness despite the weakest explanatory grounding.

75 3.1 Intrinsic disorder: seed-sensitivity in a robustness criterion

76 A proxy from the published TOP-IDP disorder-propensity scale [Campen et al., 2008] validates at
 77 $r = 0.45$ (the strongest explanation in our sweep), confirmed at the per-residue level (AUC 0.71 via
 78 a sliding window against real per-residue labels). In this case, explanatory validity and evaluation
 79 rigor are both high. The causal effect is significant and monotonically dose-responsive on all three
 80 independent seeds (Figure 2, right). However, the residue-exclusion robustness check is analogous to
 81 a faithfulness check on an explanation method. It passes on 2 of 3 seeds (36–42% of effect magnitude
 82 retained, CI excludes zero) and fails on the third (10% retained, CI crosses zero, Figure 3). The causal
 83 effect’s existence and direction are seed-stable; the specific robustness criterion is not, uniformly.

84 **Immunogenicity.** We evaluate candidate explanatory proxies across four tiers of increasing fidelity
 85 to the real target. Proxies predictive on a surrogate binding-affinity tier ($r = 0.37$ – 0.43) collapse to
 86 near-chance on the real T-cell-response endpoint ($r \leq 0.10$). A full-length-antigen proxy reaches
 87 $r = 0.38$, but organism identity alone explains 58% of the apparent signal. The same proxy’s
 88 correlation flips to -0.32 once organism identity is held out of cross-validation. This is a direct
 89 example of an apparently grounded explanation that tracks an unrelated confound. No proxy survives
 90 criterion 4, so no causal steering run was performed.

91 **Expression yield.** A net-charge-deviation proxy validates at $r = 0.31$ – 0.34 and produces a signifi-
 92 cant, dose-responsive causal effect that collapses under residue-exclusion. The explanation-validity
 93 check passes, but the faithfulness check fails.

94 4 Grounding an Ambiguous Result: A Vector-Geometry Diagnostic

95 Rather than discarding the expression-yield result as unexplained noise, we compute cosine similarity
 96 between its steering vector and every other target’s, layer by layer. We find substantial overlap with

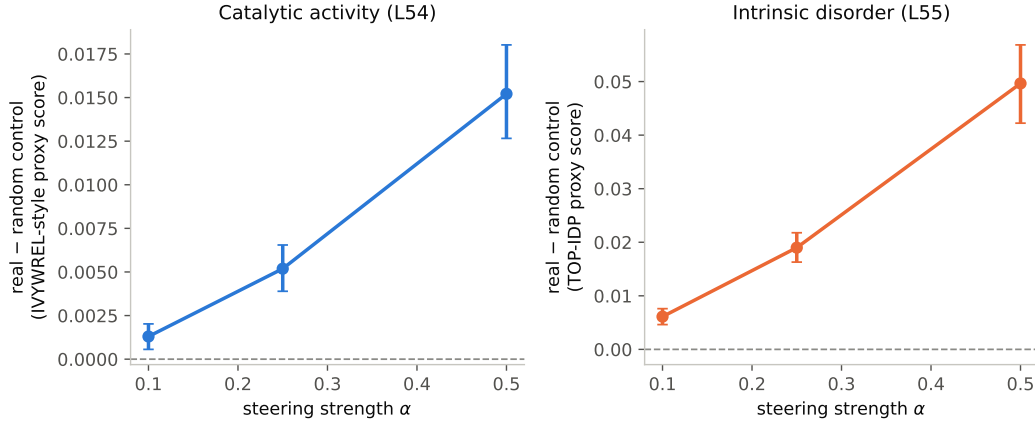


Figure 2: Dose-response for the two targets passing criterion 2. Both show a clean, monotonic increase with 95% bootstrap CIs bounded away from zero at every point.

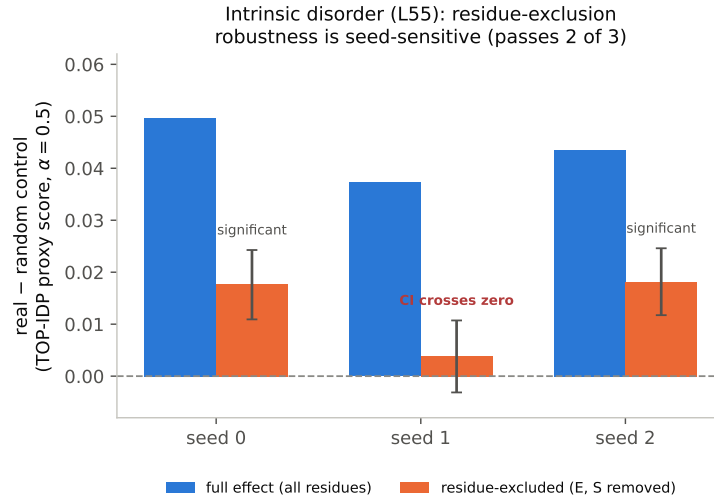


Figure 3: The intrinsic-disorder target’s causal effect (blue) and the same effect after the two dominant substituted residues are excluded (orange, 95% bootstrap CIs), across three independent seeds. The underlying causal effect is stable across all three; the residue-exclusion faithfulness check is significant on seeds 0 and 2 but its CI crosses zero on seed 1. The causal effect is unchanged, but the robustness criterion produces three different verdicts.

97 the intrinsic-disorder target’s vector: +0.38 on the full 33-layer concatenation, rising to +0.56–0.67
 98 at the three deepest layers. This is consistent with the independently documented biology because
 99 disordered protein regions are known to reduce solubility and expression yield. It also grounds
 100 the result causally: expression yield’s apparent effect is best understood as a partial, noisier echo
 101 of disorder’s already-validated direction, filtered through a proxy more prone to compositional
 102 collapse, not independent evidence of a sixth causally steerable property. This diagnostic (compare
 103 a new intervention’s direction against every other validated direction in the same study) is a cheap,
 104 generalizable faithfulness check for any study running multiple related causal interventions.

105 **A caught faithfulness-evaluation artifact.** As a mechanistic follow-up on thermostability, we
 106 tested whether restricting steering to the 5 layers previously identified as causally necessary preserves
 107 the effect of steering all 33. A first evaluation selected its best-performing α from the full tested
 108 range, including $\alpha \geq 1.0$, and concluded the 5-layer subset matched or exceeded the full intervention.
 109 This was an artifact of the evaluation, not a real result. At that α , the 33-layer intervention had

110 already fully collapsed into degenerate output, leaving zero non-degenerate evaluation sequences.
111 The 5-layer subset had not yet collapsed. This evaluation problem is directly analogous to comparing
112 two explanation methods at operating points where one has already failed. Restricting the comparison
113 to a range where neither arm has collapsed and rerunning end to end gives the honest result: the
114 5-layer subset produces a real, residue-robust effect, retaining only 43–59% of the full effect size.

115 The binding-affinity mechanistic account (§3) is likewise grounded, not asserted. The mean per-
116 residue compositional distance between this target’s low/high vector-building groups is 0.003, versus
117 0.02–0.05 for the three cross-protein targets in our sweep. This provides a measurable reason that the
118 causal construction had little signal to use, independent of the explanation’s own validity.

119 5 Related Work

120 Huang et al. [2025] introduce difference-of-means activation steering for protein language models
121 and validate it causally on thermostability; we extend their intervention to five new properties under
122 a pre-registered protocol and test, to our knowledge for the first time, whether explanatory (proxy)
123 validity predicts causal success across multiple unrelated properties. Vig et al. [2021] identify the
124 attention head we redo as a genuine causal ablation test, extended here to a full 480-head sweep
125 with zero correlational information in the ranking. Adams et al. [2025] train sparse autoencoders on
126 ESM-2’s residual stream and use a purely correlational, post-hoc linear probe to show that known
127 determinants of thermostability and subcellular localization are identifiable in SAE feature space;
128 their one steering experiment is a coherence check (does steering change fewer residues than a
129 random direction), not a validation against a real causal ground truth, and their discussion states
130 plainly that “investigating how to steer the model in useful ways remains an open direction for future
131 work.” Our work answers a version of that question directly: whether an interpretability method’s
132 correlational validity, evaluated rigorously by the field’s own standards, predicts the causal usefulness
133 of an intervention built on it.

134 6 Discussion

135 **What “causal” means here.** Every result in this paper is causal about the model’s activation space.
136 It reflects a real intervention on the forward pass, evaluated against a matched-norm random-direction
137 control and a residue-exclusion faithfulness check on generated sequences. It is not validated against
138 a real biological assay. A PASS means “causally steers a validated compositional explanation,” not
139 “causally engineers the underlying biological property.” This distinction matters when claiming that
140 an interpretability method’s causal usefulness has been established.

141 **Limitations.** All experiments use one foundation model, ESM2-650M, at one scale. We did not
142 test whether the findings hold at other scales, so it remains unclear whether the mismatch between
143 explanatory validity and causal effect is specific to this model or more general. All proxies are
144 compositional heuristics chosen for interpretability and speed rather than learned attribution methods.
145 A stronger explanation method might show a different relationship to causal effect.

146 Each target’s final proxy was selected as the best of several candidates scored on the same held-out
147 split used for its reported validation r . We did not use a separate select/test split. This choice inflates
148 every reported proxy- r , so the absolute values should be read as upper bounds. It does not obviously
149 overturn the ordinal mismatch on which the central claim rests. We also ran many significance
150 tests without a multiple-comparisons correction. Requiring multi-alpha and multi-seed agreement
151 for any PASS partially mitigates this issue, but individual borderline criteria remain close to the
152 noise floor. We checked seed robustness for only two of five targets and did not run structural or
153 functional plausibility checks on generated sequences. A PASS therefore does not rule out physically
154 or biologically implausible proteins.

155 7 Conclusion

156 Correlational validity does not predict causal effect when a downstream intervention built on it is
157 tested at two levels of the same foundation model class. This holds both for an attention head’s
158 alignment with a known biological signal and for a scoring proxy’s correlation with real experimental

159 labels. For interpretability methods intended to guide scientific discovery or trustworthy intervention
160 in biology, this argues for validating explanations against the causal interventions they are meant to
161 support, not treating correlational or attribution-based salience as self-certifying evidence of causal
162 usefulness.

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